The differences between arrays and ArrayList/Vector :

1. **Definition**:
    - **Array**: An array is a fixed-size data structure that stores elements of the same data type in contiguous memory locations.
    - **ArrayList/Vector**: ArrayList and Vector are dynamic arrays that can dynamically grow or shrink in size. They are part of the Java Collections Framework and provide resizable arrays.
2. **Size**:
    - **Array**: Arrays have a fixed size, meaning you need to specify the size when you declare them, and it cannot be changed later.
    - **ArrayList/Vector**: ArrayList and Vector can grow or shrink dynamically, allowing you to add or remove elements without worrying about the size.
3. **Type**:
    - **Array**: Arrays can only store elements of the same data type. Once declared, the data type of an array cannot be changed.
    - **ArrayList/Vector**: ArrayList and Vector can store elements of different data types. They can hold objects of any class and automatically handle resizing.
4. **Flexibility**:
    - **Array**: Arrays provide fast access to elements using index-based access. However, they cannot change size dynamically, and once created, their size cannot be altered.
    - **ArrayList/Vector**: ArrayList and Vector provide more flexibility as they can grow or shrink dynamically. They also offer various methods for adding, removing, and manipulating elements.
5. **Performance**:
    - **Array**: Arrays generally offer better performance in terms of access time since they store elements in contiguous memory locations. However, resizing arrays can be inefficient.
    - **ArrayList/Vector**: ArrayList and Vector provide better flexibility but may have slightly slower performance due to their dynamic resizing nature.
6. **Usage**:
    - **Array**: Arrays are suitable when the size is known in advance and does not change frequently.

- **ArrayList/Vector**: ArrayList and Vector are preferred when the size of the collection varies dynamically or when you need to perform frequent insertions and deletions.

Example 1

```
import java.util.ArrayList;

public class ALexample1 {
    public static void main(String[] args) {
        // Easy level example: Adding and printing elements in ArrayList

        // Creating an ArrayList
        ArrayList<String> colors = new ArrayList<>();

        // Adding elements to the ArrayList
        colors.add("Red");
        colors.add("Green");
        colors.add("Blue");

        // Printing elements of the ArrayList
        System.out.println("Colors in the ArrayList:");
        for (String color : colors) {
            System.out.println(color);
        }
    }
}
```

Example 2

```
import java.util.ArrayList;

public class ALexample2 {
    public static void main(String[] args) {
        // Medium level example: Adding, removing, and updating elements
in ArrayList

        // Creating an ArrayList
        ArrayList<Integer> numbers = new ArrayList<>();

        // Adding elements to the ArrayList
        numbers.add(10);
        numbers.add(20);
        numbers.add(30);

        // Printing elements of the ArrayList
        System.out.println("Numbers in the ArrayList:");
        for (Integer number : numbers) {
            System.out.println(number);
        }

        // Removing an element from the ArrayList
        numbers.remove(1);
```

```
        // Updating an element in the ArrayList
        numbers.set(0, 100);

        // Printing updated elements of the ArrayList
        System.out.println("Updated numbers in the ArrayList:");
        for (Integer number : numbers) {
            System.out.println(number);
        }
    }
}
```

Example 3

```
import java.util.ArrayList;

class Product {
    private int id;
    private String name;
    private double price;

    public Product(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }


    @Override
    public String toString() {
        return "Product{" +
                "id=" + id +
                ", name='" + name + '\'' +
                ", price=" + price +
                '}';
    }
}

public class ProductArrayListExample {
    public static void main(String[] args) {
        // Creating an ArrayList of Product objects
        ArrayList<Product> products = new ArrayList<>();

        // Adding products to the ArrayList
        products.add(new Product(1, "Laptop", 999.99));
        products.add(new Product(2, "Phone", 599.99));
        products.add(new Product(3, "Tablet", 299.99));

        // Displaying all products
        System.out.println("All Products:");
        for (Product product : products) {
            System.out.println(product);
        }

        // Removing a product from the ArrayList
        products.remove(1); // Removing the product at index 1 (Phone)
```

```
        // Displaying all products after removal
        System.out.println("\nProducts after removing Phone:");
        for (Product product : products) {
            System.out.println(product);
        }

        // Updating a product in the ArrayList
        Product updatedProduct = new Product(3, "Smartwatch", 199.99);
        products.set(1, updatedProduct); // Updating the product at index
1 (Tablet)

        // Displaying all products after update
        System.out.println("\nProducts after updating Tablet to
Smartwatch:");
        for (Product product : products) {
            System.out.println(product);
        }
    }
}
```